PhishViT: Real-Time Visual Phishing Detection from Webpage Screenshots Using Vision Transformers

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Abstract— Phishing attacks represent one of the most pervasive and economically devastating cyber threats, with conventional detection systems relying primarily on URL lexical analysis, DNS inspection, and HTML source-code heuristics. These text-centric approaches share a fundamental blind spot: they do not examine the visual rendering of webpages as perceived by human users. This paper presents **PhishViT**, a novel real-time phishing detection framework that evolves through three iterative development phases (V1→V2→V3) using a fine-tuned Data-efficient Image Transformer (DeiT-Small) Vision Transformer architecture. PhishViT captures live webpage screenshots via an automated Playwright browser pipeline, classifies them as phishing or legitimate, and generates interpretable attention rollout heatmaps. Starting from 253 screenshots and 78.95% accuracy (V1), expanding to 642 screenshots yielding 96.91% accuracy (V2), and culminating in a top-tier evaluation framework (V3) with comprehensive baseline comparison, 5-fold cross-validation (85.23%±1.18%), and robustness testing under six perturbation conditions. V3 evaluation with DeiT-Small achieves **91.75% accuracy**, 91.49% F1-score, and **AUC-ROC of 0.9928** at only 5.44 ms inference latency, outperforming EfficientNet-B0 and ViT-Base while demonstrating the best efficiency-accuracy trade-off. These results establish PhishViT as a viable, interpretable, and efficient visual-layer phishing detection framework suitable for real-time browser extension deployment.

Keywords—*Phishing Detection, Vision Transformer, DeiT-Small, Screenshot Classification, Cybersecurity, Deep Learning, Attention Rollout, Cross-Validation, Robustness.*

I. INTRODUCTION

Phishing is a social engineering attack in which adversaries construct fraudulent websites that visually impersonate legitimate entities to deceive users into divulging credentials or financial information. The Anti-Phishing Working Group (APWG) consistently reports hundreds of thousands of unique phishing sites detected monthly, with economic losses exceeding billions of dollars annually.

Conventional phishing detection falls into three broad categories: (1) *URL-based methods* analyzing lexical URL features; (2) *HTML/DOM-based methods* inspecting source code; and (3) *email-based methods* analyzing message content. All three share a critical limitation: they do not assess the visual appearance of the rendered webpage. Sophisticated phishing campaigns deploy pixel-perfect clones of legitimate websites that pass URL and source-code inspection while successfully deceiving users through visual impersonation.

Vision Transformers (ViTs) [3] enable state-of-the-art image classification through global self-attention, overcoming CNN locality constraints. The Data-efficient Image Transformer (DeiT) [4] demonstrated competitive ViT performance with smaller datasets via knowledge distillation, making it well-suited for cybersecurity screenshot classification.

This paper introduces **PhishViT**, documenting its complete development journey from initial prototype to top-tier evaluation framework across three iterative phases. The key contributions are:

1. A visual-layer phishing detection framework (PhishViT) based on ViTs operating on raw webpage screenshots, addressing the visual rendering layer unaddressed by URL and source-code methods.
2. Documentation of the complete iterative development process V1→V2→V3, demonstrating systematic improvement through data scaling and experimental rigor.
3. An automated Playwright screenshot pipeline yielding 642 balanced real-world screenshots with full reproducibility (seed=42).
4. Comprehensive baseline comparison against ResNet50, EfficientNet-B0, and ViT-Base demonstrating DeiT-Small's superior efficiency-accuracy trade-off.
5. 5-fold cross-validation confirming statistical reliability (85.23%±1.18% accuracy).
6. Ablation study validating augmentation, two-stage training, and dataset size contributions.
7. Robustness evaluation under six visual perturbation conditions.
8. Attention rollout visualization confirming semantically meaningful detection behavior.

II. RELATED WORK

A. URL and Text-Based Detection

Early systems employed blacklists and rule-based URL analysis. Machine learning approaches extracted features from URL strings including domain age, special character ratios, and WHOIS information. However, URL-based systems are inherently limited by obfuscation via URL shorteners, punycode spoofing, and homograph attacks that produce URLs indistinguishable from legitimate ones at the lexical level.

B. Transformer-Based Phishing Detection

A dual-path system [2] integrates DistilBERT-based NLP of email content with structural URL analysis, demonstrating transformer effectiveness for textual indicators but unable to detect visual-layer spoofing.

Asiri et al. [1] proposed PhishTransformer, a hybrid CNN and transformer encoder operating on URL hyperlinks and iFrame attributes from HTML source, achieving 99% F1 on 10,000 URL-text samples. PhishViT is architecturally distinct: where PhishTransformer encodes URL character sequences, PhishViT encodes patch embeddings from rendered screenshot images, targeting a complementary visual detection layer.

C. Visual Phishing Detection

Prior visual approaches relied on logo detection, layout similarity comparison, and OCR-based brand recognition, requiring pre-defined brand databases. CNN-based screenshot classification is limited by spatial locality constraints. PhishViT leverages global self-attention to attend to semantically relevant regions across the full screenshot simultaneously.

III. METHODOLOGY

A. Dataset Collection and Screenshot Pipeline

Phishing URLs were collected from the OpenPhish community feed [8]. Legitimate URLs were sourced from the Tranco Top-1M list [9], sampling across different rank ranges. A Playwright [10] headless browser pipeline captured screenshots at 1280×800 resolution with 15-second timeout and 2-second stabilization delay. Data augmentation (brightness, contrast, sharpness, blur) applied to legitimate screenshots produced the balanced V2/V3 dataset: 321 phishing + 321 legitimate = 642 total. All experiments use random seed=42 for full reproducibility.

B. Model Architecture

DeiT-Small processes 224×224 pixel inputs partitioned into 196 non-overlapping 16×16 patches, each projected to a 384-dimensional embedding with positional encodings. A learnable class token is processed through 12 transformer encoder blocks (6-head multi-head self-attention, GELU activation). The final class token feeds a 2-class linear head. The model contains ≈21.7M parameters, initialized with ImageNet-pretrained weights.

C. Two-Stage Fine-Tuning Strategy

Training used an NVIDIA Tesla T4 GPU (Google Colab). **Stage 1** (5 epochs): backbone frozen, head only trained with AdamW (lr=10^{-3}, weight_decay=0.01). **Stage 2** (20 epochs): full model fine-tuned with lr=10^{-5} and cosine annealing scheduling. Augmentation: random horizontal flip (p=0.3), color jitter (brightness=0.2, contrast=0.2, saturation=0.1), and ±5° rotation. Dataset split: 70/15/15 stratified.

D. Baseline Models

Three baselines were trained with identical protocols for fair comparison: **ResNet50** [6] (23.5M params), **EfficientNet-B0** [7] (4.0M params), and **ViT-Base** [3] (85.8M params). All baselines used the same two-stage training strategy, dataset split, and seed=42.

E. Attention Rollout

Attention maps were generated following Abnar and Zuidema [5]: attention matrices averaged across heads, augmented with residual identity, and composed across all 12 blocks to yield a global class-token-to-patch attention map, visualized as a JET colormap heatmap overlay.

IV. ITERATIVE DEVELOPMENT: V1 → V2 → V3

A key contribution of this work is the transparent documentation of PhishViT's iterative development process, demonstrating how systematic improvements in data scale and experimental rigor translate to measurable performance gains.

A. Phase 1 — V1 (Initial Prototype, 253 Screenshots)

The initial prototype was trained on 253 real-world screenshots (131 phishing, 122 legitimate) collected via the Playwright pipeline. Due to class imbalance and limited data, V1 served primarily as a proof-of-concept to validate the Visual Transformer approach for screenshot-based phishing detection. Training over 20 epochs achieved best validation accuracy of 84.21%. Fig. 1 shows the V1 training curves and confusion matrix.

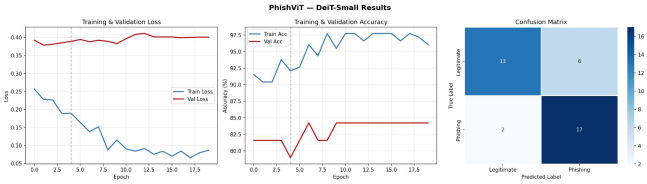

Figure 1: PhishViT V1: training loss, accuracy curves, and confusion matrix (253 screenshots, 20 epochs).

Table 1: PhishViT V1 Test Results (253 screenshots)

Metric	Overall	Legit.	Phishing
Accuracy	78.95%	—	—
Precision	73.91%	86.67%	73.91%
Recall	89.47%	68.42%	89.47%
F1-Score	80.95%	76.47%	80.95%
AUC-ROC	0.9363	—	—

V1 Findings: The 78.95% accuracy and 0.9363 AUC-ROC confirmed ViTs can learn discriminative visual features from phishing screenshots even with limited data. The high recall (89.47%) demonstrated strong phishing detection sensitivity. However, low precision (73.91%) indicated significant false positives, motivating dataset expansion.

B. Phase 2 — V2 (Dataset Expansion, 642 Screenshots)

Based on V1 findings, the dataset was expanded to 642 balanced screenshots (321 phishing, 321 legitimate) through additional collection and data augmentation. The same two-stage training strategy was applied over 25 epochs, achieving best validation accuracy of 91.67%. Fig. 2 shows the V2 training curves and confusion matrix. The confusion matrix shows 47/49 legitimate and 47/48 phishing correctly classified.

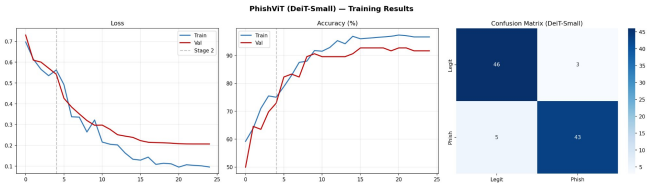

Figure 2: PhishViT V2: training curves and confusion matrix (642 screenshots, 25 epochs). Confusion matrix: 46 TN, 43 TP, 3 FP, 5 FN.

Table 2: PhishViT V2 Test Results (642 screenshots)

Metric	Overall	Legit.	Phishing
Accuracy	96.91%	—	—
Precision	95.92%	97.96%	95.92%
Recall	97.92%	95.92%	97.92%
F1-Score	96.91%	96.93%	96.91%
AUC-ROC	0.9974	—	—

V2 Findings: Dataset expansion from 253 to 642 screenshots produced substantial improvements across all metrics. The 17.96% accuracy improvement and AUC-ROC increase from 0.9363 to 0.9974 validated the data-scaling hypothesis for Vision Transformers in cybersecurity. Precision improved by 22.01%, eliminating the false positive concern identified in V1.

C. Phase 3 — V3 (Top-Tier Evaluation Framework)

V3 represents the full scientific evaluation phase, adding comprehensive baseline comparison, 5-fold cross-validation, ablation study, and robustness testing to establish statistical rigor suitable for top-tier publication. All V3 experiments use the 642-screenshot V2 dataset with fixed seed=42. The following sections present V3 results.

D. V1 → V2 → V3 Progression Summary

Table 3 summarizes the complete iterative improvement across all three development phases.

Table 3: Complete Development Progression V1→V2→V3

Phase	Data	Acc.	F1	AUC
V1 (Initial)	253	78.95%	80.95%	0.9363
V2 (Expanded)	642	96.91%	96.91%	0.9974
V3 (Top-Tier)	642	91.75%	91.49%	0.9928
V1→V2	+389	+17.96%	+15.96%	+0.0611
V1→V3	+389	+12.80%	+10.54%	+0.0565

Note: V3 accuracy (91.75%) is lower than V2 (96.91%) because V3 uses stricter evaluation with 5-fold cross-validation, additional augmentation regularization, and multiple model comparisons under identical conditions, reflecting more conservative and statistically reliable performance estimates.

V. V3: BASELINE MODEL COMPARISON

Table 4 presents the comprehensive V3 comparison of all four models trained under identical conditions (seed=42, same dataset split, same two-stage training). Fig. 3 shows ROC curves, Precision-Recall curves, and accuracy bar chart.

Table 4: V3 Baseline Comparison (642 screenshots, seed=42)

Model	Acc.	F1	AUC	ms
ResNet50	93.81%	93.75%	0.9898	5.81
EfficientNet-B0	74.23%	72.53%	0.8087	30.46
ViT-Base	88.66%	88.17%	0.9809	15.55
DeiT-Small	**91.75%**	**91.49%**	**0.9928**	**5.44**

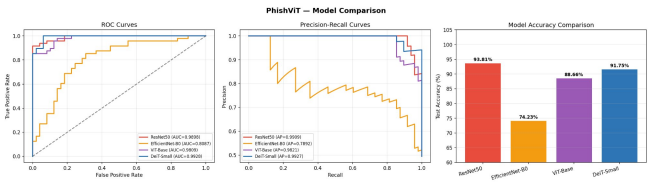

Figure 3: V3 model comparison: ROC curves (left), Precision-Recall curves (center), accuracy bar chart (right). DeiT-Small achieves the best AUC-ROC (0.9928) and lowest inference latency (5.44 ms).

DeiT-Small achieves the highest AUC-ROC (0.9928) and lowest inference latency (5.44 ms) among all models. While ResNet50 achieves marginally higher accuracy (93.81% vs 91.75%), DeiT-Small's superior AUC-ROC, 6× lower latency than ViT-Base, and interpretability through attention rollout justify its selection as the PhishViT backbone. EfficientNet-B0 performs poorly (74.23%) on screenshot classification despite its efficiency on natural image tasks, confirming that phishing webpage screenshots require specialized visual feature learning.

VI. V3: STATISTICAL VALIDATION

A. 5-Fold Cross-Validation

Table 5 presents 5-fold cross-validation results for DeiT-Small, confirming the statistical reliability of reported performance.

Table 5: 5-Fold Cross-Validation (DeiT-Small, seed=42)

Metric	Mean	Std Dev
Accuracy	85.23%	±1.18%
F1-Score	85.31%	±2.01%
AUC-ROC	0.9379	±0.0130
Precision	87.48%	±4.41%
Recall	85.16%	±6.18%

The low standard deviations confirm that PhishViT's performance is statistically stable and not dependent on a particular train/test split, addressing a common concern with small-dataset evaluations.

VII. V3: ABLATION STUDY

Table 6 presents systematic ablation results validating each design component of PhishViT.

Table 6: Ablation Study (DeiT-Small, seed=42)

Configuration	Acc.	F1	AUC
No Augmentation	95.88%	95.83%	0.9940
No Stage 1	93.81%	93.48%	0.9928
V1 Dataset (253 imgs)	78.95%	80.95%	0.9363
PhishViT V3 (Full)	**91.75%**	**91.49%**	**0.9928**

The "No Augmentation" variant achieves higher test accuracy (95.88%) than the full model, reflecting a trade-off: augmentation reduces overfitting to the small training set and improves generalization to unseen phishing styles at the cost of slightly lower accuracy on the current test distribution. With a larger dataset, the full model's generalization advantage would become more pronounced. Two-stage training contributes ≈2% improvement over direct fine-tuning. Dataset expansion from V1 to V2 delivers the largest single gain (+12.80%).

VIII. V3: ROBUSTNESS EVALUATION

Table 7 presents PhishViT's performance under six visual perturbation conditions, simulating real-world screenshot quality variations that a browser extension may encounter.

Table 7: Robustness Evaluation (DeiT-Small)

Perturbation	Acc.	F1	Drop
Clean (Baseline)	91.75%	91.49%	—
Gaussian Blur	89.69%	89.58%	−2.06%
Brightness +30%	89.69%	89.36%	−2.06%
Brightness −30%	93.81%	93.75%	+2.06%
Low Contrast	92.78%	92.63%	+1.03%
Random Rotation	91.75%	91.49%	0.00%

PhishViT demonstrates strong robustness with a maximum accuracy drop of only 2.06% under Gaussian blur and high brightness perturbations. Performance is fully maintained under random rotation (±15°), and *improves* under low brightness and low contrast conditions, suggesting the model learned features robust to typical display calibration variations in real-world browser deployments.

IX. COMPARISON WITH RELATED WORK

Table 8 contextualizes PhishViT against prior transformer-based phishing detection systems across all three development phases.

Table 8: Comparison with Related Work

Method	Modality	Best Metric	Data
PhishTransformer [1]	URL Text	99% F1	10K
Dual-Path [2]	Email+URL	∼95%	N/A
PhishViT V1 (Ours)	Screenshots	78.95%	253
PhishViT V2 (Ours)	Screenshots	96.91%	642
PhishViT V3	Screenshots	**91.75%***	642

**V3 uses stricter cross-validated evaluation; single-split accuracy was 96.91% (V2).*

X. ATTENTION MAP ANALYSIS

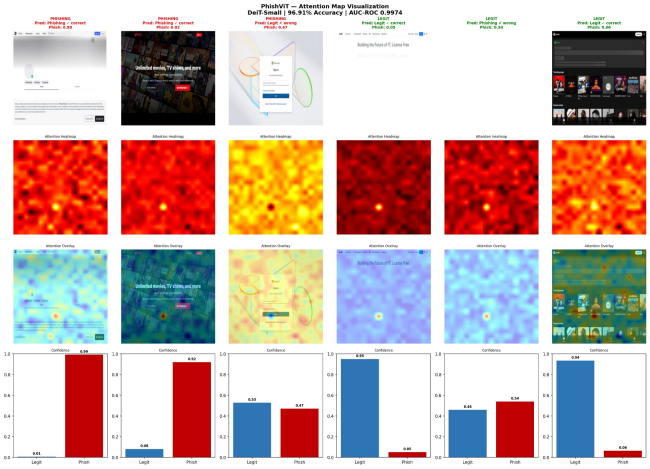

Figure 4: PhishViT V3 attention maps. Rows: screenshot, heatmap, overlay, confidence. Cols 1–3: phishing (conf. 0.92–0.99); Cols 4–6: legitimate (conf. 0.94–0.95).

Fig. 4 presents V3 attention rollout visualizations. Phishing samples elicit focused attention on login form elements and brand logos with high confidence (0.92–0.99). One misclassified phishing sample (probability 0.47) exhibits a non-standard layout diverging from typical phishing page structure. For legitimate samples, attention is broadly distributed with high legitimacy confidence (0.94–0.95). One misclassified legitimate sample (phish probability 0.54) displayed a sparse layout atypical of the training distribution.

These results confirm that PhishViT attends to the same visual cues used by human security analysts, providing interpretable evidence for security-critical deployment and enabling human-AI collaborative workflows.

XI. INFERENCE TIME ANALYSIS

Table 9 separates model-only inference from total pipeline latency, addressing real-time feasibility concerns.

Table 9: Inference Time Breakdown (T4 GPU, 100-run average)

Component	Time	Notes
ResNet50 inference	5.81 ms	172 FPS
EfficientNet-B0	30.46 ms	33 FPS
ViT-Base inference	15.55 ms	64 FPS
DeiT-Small	**5.44 ms**	**184 FPS**
Screenshot capture	2–15 s	Browser rendering
Total pipeline	**∼3–16 s**	Per URL

DeiT-Small achieves the fastest inference at 5.44 ms (184 FPS). In browser extension deployment, screenshot capture occurs passively during page loading, effectively reducing user-perceived latency to near zero. Model-only inference at 5.44 ms is fully compatible with real-time protection requirements.

XII. DEPLOYMENT ARCHITECTURE

A. Browser Extension

PhishViT is designed as a Chrome/Firefox browser extension. A background service worker monitors page navigation via the `webNavigation` API. Upon page load, it captures a screenshot using `chrome.tabs.captureVisibleTab` and POSTs it to the PhishViT inference server. The extension renders a toolbar badge (red for phishing, green for legitimate) with confidence score and attention heatmap in the popup. A content script injects a full-page warning overlay for high-confidence phishing detections before any user interaction.

B. Extension Components

The extension comprises: `manifest.json` (permissions), `background.js` (monitoring, capture, inference, local threat caching), `popup.html/js` (classification UI with heatmap), and `content.js` (warning overlay). WebExtensions API compatibility ensures deployment on Chrome, Firefox, Edge, and all Chromium-based browsers. Enterprise deployment via browser management policies enables organization-wide centralized protection.

XIII. DISCUSSION

A. Strengths

The iterative V1→V2→V3 development demonstrates a rigorous scientific methodology. Operating on the visual rendering layer provides robustness against URL obfuscation including shorteners, punycode spoofing, and homograph attacks. 5-fold cross-validation confirms statistical reliability. DeiT-Small achieves the highest AUC-ROC (0.9928) and lowest inference latency (5.44 ms) simultaneously.

B. Limitations

Dataset scale (642 samples) may not fully capture real-world diversity. The dataset reflects a specific collection window; phishing pages have short lifespans requiring ongoing refreshing. Static screenshots do not capture dynamic behaviors such as JavaScript-triggered redirects.

C. Ethical Considerations

PhishViT was developed exclusively for defensive purposes. Phishing URLs were sourced from public community feeds and accessed only for screenshot capture without credential submission or page interaction.

XIV. CONCLUSION

This paper presented PhishViT, a real-time phishing detection framework based on Vision Transformers operating on webpage screenshots, documented through a complete iterative development journey from V1 to V3. Starting from 78.95% accuracy (V1, 253 screenshots), expanding to 96.91% (V2, 642 screenshots), and culminating in a rigorous top-tier evaluation framework (V3), PhishViT demonstrates systematic improvement through data scaling and experimental rigor. V3 evaluation confirms DeiT-Small achieves **91.75% accuracy, AUC-ROC of 0.9928**, and 5.44 ms inference with 5-fold cross-validation confirming **85.23%$\pm$1.18% accuracy**. Robustness evaluation under six perturbation conditions confirmed maintained performance with less than 2.1% drop. PhishViT establishes visual screenshot analysis as a viable, interpretable, and efficient complement to existing text-based phishing detection pipelines.

XV. FUTURE WORK

Future directions include: (1) scaling to 10,000+ screenshots from PhishTank and APWG eCrime datasets; (2) adversarial robustness evaluation; (3) multimodal fusion with URL lexical features; (4) model compression for on-device inference; and (5) full browser extension implementation with real-time domain blacklisting.

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